Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building data systems on **SQL Server** and Azure SQL, writing queries, schemas, and SQL transformations, developing **C#/.NET** services, and maintaining production software with Git, testing, and 24/7 reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized relational schemas, tables, and indexes on **SQL Server** and Azure SQL with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through query tuning.
- Built SQL transformations and recurring integration jobs across SQL Server, PostgreSQL, and cloud storage with validation and reconciliation, maintaining 5K+ monthly jobs at 99%+ **production reliability** for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed and maintained a production Python and Flask application with normalized **MySQL** tables on AWS RDS, serving 150+ instructors and 10K+ daily requests with stable performance at peak load.
- Built concurrent batch processing for 1K+ uploaded files per run with fault-tolerant execution and error handling, shipping changes through Git and GitHub Actions **CI/CD** and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Supported **production software** across Windows, Linux, and macOS with nightly automated builds, installs, and 20+ unit and integration test suites, coordinating with US and India teams on parallel builds.
- Troubleshot and optimized large-scale integrations with **SQL** indexing, caching, and object-oriented Singleton and Factory designs, improving response times 30% and reducing technical debt 20% in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **C# ASP.NET Core** Web API microservices with Entity Framework on Azure SQL for resource allocation, billing, and reporting, synchronizing 2M+ records across locations in real time.
- Modeled structured data in Azure SQL for users, resources, invoices, and usage reports, and automated Jenkins **CI/CD** with unit, integration, and end-to-end tests, reaching 95%+ build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **C# .NET** Web APIs for bookings, pricing, and payments with transactional locking against double-booked time slots, sustaining sub-200ms latency and 40% faster confirmations across 50K+ records.
- Automated refund and dispute processing through idempotent payment webhooks, cutting manual handling 70%, and built hourly Databricks **data aggregation** on peak hours and service demand for client reporting.

Skills

Languages: C#, SQL, T-SQL, Python, Java, TypeScript, JavaScript
Databases: Microsoft SQL Server, Azure SQL, PostgreSQL, MySQL, Stored Procedures, Views, Schema Design, Indexing, Query Optimization
.NET & Backend: .NET, ASP.NET Core, Web API, Entity Framework, REST APIs, Microservices, Flask, FastAPI
Version Control & DevOps: Git, GitHub, CI/CD, Jenkins, GitHub Actions, Docker, Microsoft Azure, AWS, Windows, Linux
Quality & Operations: Unit Testing, Integration Testing, Production Support, Troubleshooting, Change Management
Practices: Requirements Analysis, Agile/Scrum, Global Team Collaboration

Projects

Fab Recipe Configuration Manager | *C#, .NET 8, SQL Server, T-SQL, Entity Framework, xUnit*

- Designed a C# and SQL Server tool that applies **change-order** driven equipment configuration updates through versioned **stored procedures**, validates against reference tables, and keeps an auditable rollback history.

Chat2SQL Code | *Java, TypeScript, SQL Server, PostgreSQL, Docker*

- Extended a database workspace on Java and TypeScript supporting 10+ systems including **SQL Server**, with schema synchronization and AI-assisted query generation at 82% accuracy for faster database work.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling** algorithms in C++ with a React Gantt visualization, testing 1K+ process combinations and showing 25% lower average turnaround.